

SCIENTIFIC PUBLICATIONS / PUBLICATIONS SCIENTIFIQUES

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Journals with a reviewing committee

1. Sarbout I., Ounissi M., Milea D., Racoceanu D. (2026) From Synthetic Navigation Data to Real-World Mobility Cues: Reinforcement Learning for Sensory Substitution in Visual Impairment", *Array*, 30:100861. <https://www.sciencedirect.com/science/article/pii/S2590005626001840>.
2. Ounissi M., Sarbout I., Hugot J.-P., Martinez-Vinson C., Berrebi D., Racoceanu D. (2025). Scalable, trustworthy generative model for virtual multi-staining from H&E whole slide images. ***PLoS Comput. Biol.*** 21(10): e1013516. <https://doi.org/10.1371/journal.pcbi.1013516>
3. Zhao, C., Li, J., Zhao, Q., Bai, J., Boluda, S., Delatour, B., Stimmer, L., Racoceanu, D., Jimenez, G., & Fu, G. (2025). ADNP-15: An open-source histopathological dataset for neuritic plaque segmentation in human brain whole slide images with frequency-domain image enhancement for stain normalization. *Innovation and Research in Biomedical Engineering (IRBM)*, 46(6), Article 100913. <https://doi.org/10.1016/j.irbm.2025.100913>
4. Sarbout I, Gungor A, Ounissi M, Zaher S , Ptito M, Kupers R, Racoceanu D, Milea D. (2025). Visual Prostheses in the Era of Artificial Intelligence Technology, ***Eye and Brain***, Volume 2025:17 Pages 95—113. DOI : <https://doi.org/10.2147/EB.S524322>
5. Gungor, A., Sarbout, I., Gilbert, A. L., Hamann, S., Lebranchu, P., Hobeau, C., Gohier, P., Vignal-Clermont, C., Dumitrascu, O. M., Cohen, S. Y., Lagrèze, W. A., Feltgen, N., van der Heide, F., Lamirel, C., Jonas, J. B., Obadia, M., Racoceanu, D., & Milea, D. (2025). Artificial Intelligence-Based Detection of Central Retinal Artery Occlusion Within 4.5 Hours on Standard Fundus Photographs. ***Journal of the American Heart Association***, 14(13), e041441. <https://doi.org/10.1161/JAHA.124.041441>.
6. Marin, L.E., Zavaleta-Guzman, D.I., Gutierrez-Garcia, J.I., Racoceanu, D., Casado, F.L. (2025) Prediction of biochemical prostate cancer recurrence from any Gleason score using robust tissue structure and clinically available information. *Discover Oncology*, 16(1), art. no. 128. <https://doi.org/10.1007/s12672-025-01896-7>.
7. Kumar, P., Lacroix, M., Dupré, P., Arslan, J., Fenou, L., Orsetti, B., Le Cam, L., Racoceanu, D., Radulescu, O. (2024). Deciphering oxygen distribution and hypoxia profiles in the tumor microenvironment: a data-driven mechanistic modeling approach. *Phys Med Biol.* 2024 Jun 14; 69(12). <https://iopscience.iop.org/article/10.1088/1361-6560/ad524a>. PMID: 38815610.
8. Ingrassia, L., Boluda, S., Jimenez, G., Kar, A., Racoceanu, D., Delatour, B., Stimmer, L. (2024). Automated deep learning segmentation of neuritic plaques and neurofibrillary tangles in Alzheimer's disease brain sections using a proprietary software, *Journal of Neuropathology and Experimental Neurology*. <https://doi.org/10.1093/jnen/nlae048>.
9. Ounissi, M., Latouche, M., Racoceanu, D. (2024). PhagoStat a scalable and interpretable end-to-end framework for efficient quantification of cell phagocytosis in neurodegenerative disease studies, *Scientific Reports*, <https://doi.org/10.1038/s41598-024-56081-7>, <https://www.nature.com/articles/s41598-024-56081-7>.
10. Liu, X., Hu, W., Diao, S., Abera, D. E., Racoceanu, D., Qin, W. (2024). Multi-scale feature fusion for prediction of IDH1 mutations in glioma histopathological images, *Computer Methods and Programs in Biomedicine*, Elsevier, vol. 248, pp. 108116, doi: <https://doi.org/10.1016/j.cmpb.2024.108116>
11. Diao S., Tian Y., Hu W., Hou J., Lambo R., Zhang Z., Xie Y., Nie X., Zhang F., Racoceanu D., Qin W. (2022). Weakly supervised framework for cancer region detection of hepatocellular carcinoma in whole-slide pathologic images based on multiscale attention convolutional neural network. *American Journal of Pathology*. 192(3):553-563. doi:[10.1016/j.ajpath.2021.11.009](https://doi.org/10.1016/j.ajpath.2021.11.009).

12. Fraggetta F, L'imperio V, Ameisen D, Carvalho R, Leh S, Kiehl TR, Serbanescu M, Racoceanu D, Della Mea V, Polonia A, Zerbe N, Eloy C. (2021). Best practice recommendations for the implementation of a digital pathology workflow in the anatomic pathology laboratory by the european society of digital and integrative pathology (ESDIP). *Diagnostics*, 11(11):2167. doi:[10.3390/diagnostics11112167](https://doi.org/10.3390/diagnostics11112167).
13. Jiménez, G., Racoceanu, D. (2019). Deep Learning for Semantic Segmentation versus Classification in Computational Pathology: Application to mitosis analysis in Breast Cancer grading, *Front. Bioeng. Biotechnol.*, 21 June 2019, doi: 10.3389/fbioe.2019.00145.
14. Marin, L., Casado, F., Racoceanu, D. (2019), Classification of prostate cancer based on clinical and omic data using neural networks techniques to improve prognostic power, *European Urology Supplements* 18(1):e1783; DOI: 10.1016/S1569-9056(19)31292-8.
15. Marin, L., Casado, F., Pinto, J.A., Racoceanu, D., (2019). Classification of prostate cancer based on clinical and omics data using neural networks techniques to improve prognostic power, *Journal of Clinical Oncology* 37(15_suppl):e16569-e16569 ; DOI: 10.1200/JCO.2019.37.15_suppl.e16569.
16. Zemouri, R., Zerhouni, N., Racoceanu, D. (2019). Deep Learning in the Biomedical Applications: Recent and Future Status, *Appl. Sci.*, 9(8), 1526; doi:10.3390/app9081526.
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24. Racoceanu, D., Ameisen, D., Veillard, A., Ben Cheikh, B. Attieh, E., Brezillon, P., Yunès, J.-B., Temerson, J.-M., Toubiana, L., Verger, V., Pomerol, J.-F., Klossa, J., Lallemand, F., Constant, P., Capron, F., Guettier, C., Phan, N., Bertheau P. (2016). Towards efficient collaborative digital pathology: a pioneer initiative of the FlexMlm project, *The Diagnostic Pathology Journal*, 8: 199.
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28. Racoceanu D, & Capron F, (2016). Semantic Integrative Digital Pathology: Insights into Microsemiological Semantics and Image Analysis Scalability. *Pathobiology*, 83: 148-155.
29. Racoceanu, D., & Capron, F., (2015). Towards Semantic-Driven High-Content Image Analysis. An Operational Instantiation for Mitosis Detection in Digital Histopathology. *Computerized Medical Imaging and Graphics*, 2: 2-15. DOI: 10.1016/j.compmedimag.2014.09.004.

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37. Loménie, N., Racocceanu, D. (2012). Point set morphological filtering and semantic spatial configuration modeling: applications to microscopic image and bio-structure analysis. *Pattern Recognition*, 45(8): 2894-2911. ISSN: 0031-3203. DOI: 10.1016/j.patcog.2012.01.021.
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45. Palluat, N., Racocceanu, D., Zerhouni, N. (2006). A neuro-fuzzy monitoring system. Application to flexible production systems. *Computers in Industry, Special Issue: E-maintenance*, Ed. Elsevier Science, 57(6): 528-538.
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Edited books, symposium proceedings, scientific journals special issues

1. *Special Issue MICCAI 2020*, Purang Abolmaesumi, Anne Martel and Daniel Racocceanu (Eds), *Medical Image Analysis*, sept. 2021, ISSN: 1361-8415.
2. Martel, A.L., Abolmaesumi, P., Stoyanov, D., Mateus, D., Zuluaga, M.A., Zhou, S.K., Racocceanu, D. and Joskowicz, L. (Eds.), *Medical Image Computing and Computer Assisted Intervention – MICCAI 2020*, 23rd International Conference, Lima, Peru, October 4–8, 2020, Proceedings (7 volumes / parts), Springer: [Part I](#), [Part II](#), [Part III](#), [Part IV](#), [Part V](#), [Part VI](#), [Part VII](#).
3. Racocceanu, D., Hufnagl, P. (Eds), Special Issue - "Scalable and Efficient Imaging Technologies for Digital Pathology" (2017). CMIG - Computerized Medical Imaging and Graphics, Elsevier, vol. 61, pp.1-34.
4. Racocceanu, D., Belhomme, P., (Eds.), Special issue: "Breakthrough technologies in digital pathology", CMIG - Computerized Medical Imaging and Graphics, Elsevier, vol. 2, June 2015, pp.1, DOI: [http://dx.doi.org/10.1016/S0895-6111\(15\)00047-6](http://dx.doi.org/10.1016/S0895-6111(15)00047-6).
5. Roux, L., Racocceanu, D., (Eds.) Mitosis detection in breast cancer histological images: An ICPR 2012 contest, *Journal of Pathology Informatics*, special issue, vol. 4, issue 1, 30 May 2013, ISSN: 2153-3539.
6. Loménie, N., Racocceanu, D., Gouaillard, A., (Eds.), *Advances in Bio-Imaging: From Physics to Signal Understanding Issues, State-of-the-Art and Challenges*, Springer Series: *Advances in Intelligent and Soft Computing*, vol. 120, 2012, XXII, 246 p., ISBN 978-3-642-25546-5, January 9, 2012.
7. Leow, W. K., Feng, D., Li, H., Racocceanu, D. (Eds.), *SinFra'09 - Singaporean-French IPAL Symposium*, Proceedings of the symposium with the selected publications, World Scientific, ISBN: 978-981-4277-55-6(CD) / 981-4277-55-X(CD), 18–20 February 2009, Fusionopolis & Institute for Mathematical Sciences, Singapore.
8. Nakai, T., Racocceanu, D. (Eds.), Special Issue on ONtology and COntext related MEdical image Distributed Intelligent Access, International Symposium of the 26th Annual Meeting of Japanese Society for Medical Imaging Technology (JAMIT), Tsukuba International Convention Center, 20 July 2007, Tsukuba, Japan, *Medical Imaging Technology*, vol.25 No.5, Nov. 2007.

Book Chapters

1. Popovici, V., Racocceanu, D. (2025). From histopathology images to molecular characterisation of tumours: The artificial intelligence path, in *Recent Advances in Histopathology*, Vol 27 – Ed. Fred T Bosman and Dr. Ivan Damjanov. Jaypee Brothers Medical Publishers. [\[DOI\]](#)
2. Jiménez, G., Racocceanu, D. (2023). Computational Pathology for Brain Disorders. In: Colliot, O. (eds) *Machine Learning for Brain Disorders*. Neuromethods, vol 197, pp. 533-572 (40 pages), Humana Press, New York, NY. [\[DOI\]](#)
3. Racocceanu D, Ounissi M, Kergosien Y. (2022) Explicabilité en Intelligence Artificielle ; vers une IA Responsable. *Instanciation dans le domaine de la santé. Techniques de l'Ingénieur*. Published online December 10, 2022. doi:10.51257/a-v1-h5030 – 22 pages.

4. Zemouri, R., Racoceanu, D. (2021). Innovative Deep Learning Approach for Biomedical Data Instantiation and Visualization, in Deep Learning for Biomedical Data Analysis: Techniques, Approaches/Application, Springer.
5. Lomenie, N., Racoceanu, D. (2012). Ontology-Enhanced Vision System for New Microscopy Imaging Challenges, Advances in Bio-Imaging: From Physics to Signal Understanding Issues, vol. 120, series Advances in Intelligent and Soft Computing, pp 157-172.
6. Loménie, N., Racoceanu, D., Stamon, G. (2011). Point Set Analysis: An Image Analysis Point of View for Rapid Prototyping Technologies, in Rapid Prototyping / Book 1", Ed. M. Hoque, ISBN 978-953-307-970-7, InTech, July 2011.
7. Teodorescu, R. O., Cretu, V. I., Racoceanu, D. (2013). Parkinson's Disease Diagnosis and Prognosis Using Diffusion Tensor Medical Imaging Features Fusion, "Biomedical Engineering, Trends in Electronics, Communications and Software, Ed. Anthony N. Laskovski, ISBN: 978-953-307-475-7, InTech, Jan. 2011 (3000 downloads @ 01/2013).
8. Su, M. J., Cheng, P. H., Chen, S. J., Yang, C. Y., Yip, P. K., Racoceanu, D., Chen, H. S. (2010). Medical Image Intelligent Access Integrated with Electronic Medical Records System for Brain Degenerative Disease, Data Storage, Book edited by: Florin Balasa, ISBN: 978-953-307-063-6, Publisher: InTech, Apr. 2010, pp. 201-211.
9. Lacoste, C., Chevallet, J. P., Lim, J. H., Hoang, D. L. T., Xiong, W., Racoceanu, D., Teodorescu, R. O., Vuillenemot, N. (2007). Inter-Media Concept-Based Medical Image Indexing and Retrieval with UMLS at IPAL, in Lecture Notes in Computer Science, Evaluation of Multilingual and Multi-modal Information Retrieval, vol. 4730, pp. 694-701.

Conferences Invitee

1. Racoceanu D. (2026), EMBL-EBI "AI Reasoning with LLMs: Applications in Drug Discovery and Precision Medicine" workshop, Pfizer, Boston, USA.
2. Racoceanu D. (2026), Keynote, SIPAIM 2026, Lima, Peru.
3. Racoceanu D. (2026), AAAI Conference on Artificial Intelligence – AAAI-26, SPARTA workshop — Spatial Reasoning and Therapeutics with AI : From Omics to Imaging, Singapore.
4. Racoceanu D. (2025), Hand in hand with XAI on a biomedical imaging journey, Journées Ouvertes en Biologie, Informatique et Mathématiques (JOBIM), Bordeaux, France.
5. Racoceanu D. (2025). Explainable Artificial Intelligence (XAI): a modern Ariadne's thread in Biomedical Imaging. Supporting Discovery of new knowledge, Frugal environmental respectful Computational Approaches, Computer Vision for Drug Discovery (CVDD) workshop, IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), Nashville, TN, USA.
6. Racoceanu, D. (2024). Explainable, Hand in hand with AI on a medical imaging journey, Romanian AI days, Bucharest, Romania.
7. Racoceanu, D. (2024). Virtual staining using trustworthy and scalable pipeline, Invited Speaker, European Congress on Digital Pathology – ECDP 2024, Vilnius, Lithuania.
8. Racoceanu, D. (2024). What would Socrates say about AI? - *Guest of Honor* - Annual Conference of the National Academy of Medical Sciences (NAMS), hosted by Bombay Hospital Institute of Medical Sciences, Mumbai India.
9. Racoceanu, D. (2024). Last advances in AI and eXplainable AI, invited speaker, Soirée de Recherche à l'Hôpital Fondation Adolphe de Rothschild, Paris, France.
10. Racoceanu, D. (2024). Qvo vadis homine, AI? invited speaker, CEFIPRA Indian-French workshop on XAI for Brain pathologies, Vijayawada, Inde, 19-21 Feb. 2024.
11. Racoceanu, D. (2023). Tumor heterogeneity, tumor micro-environment analysis, modeling and simulation and sparse 3D data reconstruction in computational histopathology, Keynote speaker, MICCAI workshop on Computational Mathematics Modeling in Cancer Analysis (CMMCA2023), MICCAI - International Conference on Medical Image Computing and Computer-Assisted Intervention, Vancouver, Canada.
12. Racoceanu, D. (2023). Explicabilité en Intelligence Artificielle : challenges et opportunités en santé et en recherche biomédicale, invited speaker, Rencontres DIM C-BRAINS – Google, « IA, cerveau et cloud computing : explorer les frontières de l'innovation », 13 septembre 2023, Paris, France.
13. Racoceanu, D. (2021). Explainable IA: towards a Responsible AI, keynote speaker, SIPAIM 2021 - 17th International Symposium on Medical Information Processing and Analysis, Campinas, Brazil.

14. Racoceanu, D. (2019). Vessels detection for angiogenesis quantification deep learning & marked point process, ECDP 2019 – 15th European Congress on Digital Pathology, Warwick, UK, 10-13 April, 2019.
15. Racoceanu, D. (2019). Tumor Heterogeneity Analysis, Indexing and Simulation in Computational Pathology, Shenzhen Institute of Advanced Technology (SIAT), Shenzhen, China.
16. Racoceanu, D. (2019). Computational Pathology: A Path Ahead, Café de la Neuroinformatique, Institut du Cerveau et de la Moelle épinière (ICM), Paris, France.
17. Racoceanu, D. (2019). Artificial intelligence for biomedical image analysis applications: lessons learned from personal projects, expertise and contributions, Institut du Cerveau et de la Moelle épinière (ICM), Paris, France, 10 May 2019.
18. Racoceanu, D. (2018). Integrative Computational Histopathology with Latin America. Invited speaker, Computational Pathology Symposium, ECP 2018 - 30th European Congress of Pathology, Bilbao, Spain.
19. Racoceanu, D. (2018). A Simulation Model for Computational Histopathology issued from the study of Tumour Microenvironment. An instantiation in Breast Carcinomas, Invited speaker, 14th European Congress on Digital Pathology, Helsinki, Finland, May 29th – June 1st, 2018.
20. Racoceanu, D. (2018). Artificial Intelligence Applied to Very Large Image Analysis in Medicine (Digital Pathology), Invited Speaker, Día Mundial de las Telecomunicaciones, la Sociedad de la Información, y el Internet, Pontifical Catholic University of Peru.
21. Racoceanu, D. (2017). Integrative Computational Pathology and beyond ..., Keynote speaker, Digital Pathology and Image Analysis, San Antonio, USA, November 15-16, 2017.
22. Racoceanu, D. (2017). Integrative Digital Pathology and beyond, Invited Speaker, National University of Colombia, Bogota, Colombia, 9th of May 2017.
23. Racoceanu, D. (2016). Integrative digital pathology: semantics and scalable imaging methods, DIGITALPath Europe 2016, London, 18-19 May 2016.
24. Racoceanu, D. (2016). Are we ready to deal with traceability and massive data in Digital Pathology?, Special session on Biomarker Detection and Discovery in Histopathology Images, 2016 IEEE International Symposium on Biomedical Imaging (IEEE ISBI), Prague, Czech Republic, 13-16 April 2016.
25. Racoceanu, D., Rossi, F. (2016). Terres d'innovation Singapour, une plate-forme d'innovation et de R&D unique en Asie, Invited Speaker, Congres C.U.R.I.E., 6-9 June 2016, Deauville, France.
26. Racoceanu, D. (2015). Semantics, micro-semiology and scalability in future Digital Pathology, Digital Pathology Congress, 21-22 September 2015, Kuala Lumpur, Malaysia.
27. Racoceanu, D. (2015). Insights about scalability and big data in Digital Pathology. Recent projects and outcomes, Dublin Pathology 2015. 8th Joint Meeting of the British Division of the IAP and The Pathological Society, June 23rd, 2015, Dublin, Ireland.
28. Racoceanu, D. (2015). Semantic-Driven Image Exploration for WSI. Scalability for big data, Information Technology in Digital Pathology, AIDPATH EU, May 26th, 2015, Madrid, Spain.
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30. Racoceanu, D. (2014). Big data in microscopy, Invited Speaker, French National Center for Scientific Research (CNRS) Research Group - Technologies for Healthcare - GdR STIC-Santé, Lyon, France, 23-25 Sept. 2014.
31. Racoceanu, D. (2014). Towards the future of Digital Pathology, Invited Talk, Massachusetts General Hospital - MGH, Boston, MA, USA, 17 Sept. 2014.
32. Racoceanu, D. (2014). Digital Pathology: a semantic driven protocol for WSI exploration, Invited Speaker, Harvard Medical School, Beth Israel Deaconess Medical Center (BIDMC), Pathology Department, Boston, MA, USA, 15 Sept. 2014.
33. Racoceanu, D. (2014). Major research results in cognitive multi-scale exploration of high content images at IPAL/BMIU, Invited Speaker, Bioinformatics Institute BII - A*STAR, Singapore, 7th of February 2014.
34. Racoceanu, D. (2013). Symbolic cognitive approaches using prior knowledge and prior shapes for whole slide image understanding. Roadmap and perspectives inspired by a long-run french-singaporean collaboration, 3rd European Conference on Whole Slide Imaging and Analysis, Heidelberg, Germany, 29-30 Nov. 2013.
35. Racoceanu, D. (2013). Exploration cognitive symbolique des images biomédicales haut contenu. Application à la gradation des cancers en histopathologie et au suivi de cellules souches par recalage

dynamique entre l'analyse 2D et la synthèse 3D, Invited Speaker, LIRMM UMR CNRS – University of Montpellier II, 24 Sept. 2013, Montpellier, France.

36. Racocceanu, D. (2013). Cognitive virtual microscopy for breast cancer grading in histopathology Whole Slide Image exploration using a symbolic cognitive vision approach, Invited Speaker, Centre de Recherche des Cordeliers, UMR INSERM, 8 Apr. 2013, Paris, France.
37. Racocceanu, D. (2013). Symbolic visual approaches using prior knowledge and shapes for high content microscopic images exploration. Roadmap and perspectives inspired by a long-run french-singaporean collaboration, Invited Speaker, INRIA/CNRS-Sophia Antipolis, Ayin (INRIA) and Morpheme (INRIA/CNRS) research teams, invited talk, 18 Feb, 2013, Sophia-Antipolis, France.
38. Racocceanu, D. (2013). Approches cognitives symboliques et connaissances/formes a priori pour l'imagerie microscopique haut contenu. Perspectives à travers des projets franco-singapouriens, Invited Speaker, LTSI, French National Institute of Health and Medical Research, INSERM UMR, University Rennes 1, 14 Feb. 2013, Rennes, France.
39. Racocceanu, D. (2013). Approches visuelles sémantiques en histopathologie, Invited Speaker, Unité BioTICLA - Equipe IMAGIN' Biologie et Thérapies Innovantes des Cancers Localement Aggressifs EA 4656 de l'Université de Caen Basse-Normandie, IFR 146 ICORE Centre de Lutte Contre le Cancer François Baclesse, 12 Feb 2013, Caen, France.
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44. Racocceanu, D. (2010). Presentation of the International French Singaporean Collaboration IPAL – Image & Pervasive Access Lab – UMI CNRS, French Singaporean Entrepreneurships R&D Committee Opening Meeting, invited speaker, Organized by the French Chamber of Commerce in Singapore (FCCS) & French Embassy in Singapore, 14 January 2010, Singapore.
45. Racocceanu, D. (2009). IPAL research in ontology-driven virtual microscopy for breast cancer grading, European COST Action IC0604 Euro-Telepath, Management Committee and 6/7th Working Groups Meetings, Conference Invited Expert, 03-04 Dec. 2009, Lausanne, Switzerland.
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47. Racocceanu, D. (2008). Image Based Reasoning using Medical Knowledge to improve Diagnosis Assistance. Showcases: Early Detection of Brain stroke from brain CT, Histopathology Imaging-based Automatic Breast Cancer Grading, First International Symposium on ICT For Health - ICT4Health 2008, Keynote Lecture, February 29th -1st March 2008, Ateneo de Manila University, Manila, Philippines.
48. Racocceanu, D. (2008). Image-Based Reasoning Systems using Medical Knowledge for Diagnosis Assistance, Invited Speaker, Institute of Molecular and Cell Biology (IMCB), Biopolis, Agency for Science, Technology and Research (A*STAR), Singapore, 27 February 2008.
49. Racocceanu, D. (2005). Monitoring Approach Using Recurrent Radial Basis Function Neural Networks and Neuro-Fuzzy Systems, Invited Speaker, Department of Computer Science, School of Computing, National University of Singapore, 7 Dec. 2005.

50. Racocceanu, D., Zerhouni, N. (2005). Dynamic Monitoring Systems using Recurrent Neural Networks and Neuro-Fuzzy Systems. Nemosys Project, Invited Speaker, Artificial Intelligence Department, Tsinghua University, Beijing, China, 17 October 2005.
51. Racocceanu, D. (2002). Réseaux de Neurones Temporels. Application à la Maintenance Préventive et à la Télémaintenance, Conference Invitee, Conférence Internationale d'Ingénierie Intégrée - C2I 2002, 25-26 April 2002, Timisoara, Romania.

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1. Arslan, J., Racocceanu, D., Benke, K. K. (2023). Deep Learning Using Images of the Retina for Assessment of Severity of Neurological Dysfunction in Parkinson Disease, JAMA Ophthalmol. Published February 9, 2023. doi:10.1001/jamaophthalmol.2022.6036.

Datasets, Challenges

1. Zhao, C., Li, J., Zhao, Q., Bai, J., Boluda, S., Delatour, B., Stimmer, L., Racocceanu, D., Jimenez, G., & Fu, G. (2025). ADNP-15: An open-source histopathological dataset for neuritic plaque segmentation in human brain whole slide images with frequency-domain image enhancement for stain normalization. *Innovation and Research in Biomedical Engineering (IRBM)*, 46(6), Article 100913. <https://doi.org/10.1016/j.irbm.2025.100913>
2. Jimenez G., Kar A., Ounissi M., Ingrassia L., Boluda S., Delatour B., Stimmer L., Racocceanu D. (2022). Visual Deep Learning-Based Explanation for Neuritic Plaques Segmentation in Alzheimer's Disease Using Weakly Annotated Whole Slide Histopathological Images. In: Wang, L., Dou, Q., Fletcher, P.T., Speidel, S., Li, S. (eds) *Medical Image Computing and Computer Assisted Intervention – MICCAI 2022*. Lecture Notes in Computer Science, vol 13432. Springer, Cham. https://doi.org/10.1007/978-3-031-16434-7_33
3. Roux, L., Racocceanu, D., (Eds.) Mitosis detection in breast cancer histological images: An ICPR 2012 contest, *Journal of Pathology Informatics*, special issue, vol. 4, issue 1, 30 May 2013, ISSN: 2153-3539.

International Communications with Proceedings

1. Ostermaier J., Ruyter S., Dorent R., Racocceanu D. (2026). Disentangled Shared Representations Improve Morpho-Transcriptomic Integration, COMPAYL 2026, MICCAI 2026, Strasbourg France.
2. Ruyter S., Dorent R., Racocceanu D. (2026). Program-level generative modeling of spatial transcriptomics from tissue morphology, COMPAYL 2026, MICCAI 2026, Strasbourg France.
3. Ruyter S., Dorent R., Racocceanu D. (2026). Normalization Bias in Morpho-Transcriptomic Prediction, Medical Image and Deep Learning - MIDL 2026 Short Paper Track, Taipei, Taiwan.
4. Ruyter S., Racocceanu D. (2026). When High Correlation Isn't Enough: Rethinking Evaluation in H&E-Based Gene Expression Prediction, ECDP - European Congress on Digital Pathology, Gratz, Austria
5. Ostermaier J., Ruyter S., Racocceanu D. (2026). Cross-Scale Coherence: Evaluating Biological Consistency in Multimodal Latent Spaces, ECDP - European Congress on Digital Pathology, Austria.
6. Sarbout, I., Bodart, G., Cavallo, D., Decottignies, V., Vignal, C., Racocceanu, D., Gohier, P., Chiquet, C., Milea, D. (2026). *Data-driven structure-function model improves central visual field mapping in NAION*, North American Neuro-Ophthalmology Society - NANOS 2026 Annual Meeting, Boston, US.
7. Gungor, A., Zaher, S., Sarbout, I., Croitoru, S., Huna-Baron, R., Touzani, B., Lamirel, C., Banc, A., Racocceanu, D., Audren, F., Keane, P. A., & Milea, D. (2026). *Prospective AI-based retinal image identification of neuro-ophthalmic disorders in a tertiary emergency department*, North American Neuro-Ophthalmology Society - NANOS 2026 Annual Meeting, Boston, US.
8. Sarbout, I., Cazenave-coupet, T., Ounissi, M., Milea, D., Racocceanu, D. (2026). City of Light (COL): A City-Scale, Geo-Anchored Urban Simulator with High-Throughput Multi-Sensor Streams. AAI 2026 Singapore.
9. Ruyter S., Ounissi M., Racocceanu D. (2025). Diffusion Models for Morphology-Guided Transcriptomics: A Computational Framework, European Congress on Digital Pathology ECDP 2025, Barcelona, Spain.
10. G. Jimenez, L. Hebert-Stevens, S. Boluda, B. Delatour, L. Stimmer, D. Racocceanu (2025) Unravelling the Topographical Organization of Brain Lesions in Variants of Alzheimer's Disease Progression, Proc. SPIE 13413, Medical Imaging 2025: Digital and Computational Pathology, 134130J; <https://doi.org/10.1117/12.3047321>.
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12. Gungor, A., Sarbout, I., Hage, R., Gohier, P., Lebranchu, P., Dumitrascu, O., Vignal-Clermont, C., Chen, J.J., Rattanathamsakul, N., Lagreze, A., Feltgen, N., Obadia, M., Racocceanu, D., Milea, D., (2024). Early Detection of Central Retinal Artery Occlusion within 4.5 Hours of Visual Loss: Deep Learning Method Applied on Fundus Photographs, Congress of the European Neuro-ophthalmology Society – EUNOS 2024, Rotterdam, Netherlands *** Best Poster award – EUNOS 2024.**
13. Milea, D., Gungor, A., Sarbout, I., Hage, R., Gohier, P., Lebranchu, P., Dumitrascu, O., Vignal-Clermont, C., Chen, J.J., Rattanathamsakul, N., Obadia, M., Racocceanu, D. (2024). Diagnosis of Central Retinal Artery Occlusion within 4.5 hours after Visual Loss, using a Deep Learning Method Applied on Fundus Images, North American Neuro-Ophthalmology Society 50th Annual Meeting – NANOS 2024, March 2nd - 7th, 2024, USA.
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15. Jimenez Garay, G., Mas, P., Kar, A., Ingrassia, L., Boluda, S., Benoit Delatour, B., Lev Stimmer, Daniel Racocceanu (2023). A meta-graph approach for analyzing whole slide histopathological images of human brain tissue with Alzheimer disease biomarkers, SPIE Medical Imaging, San Diego, USA.
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17. Arslan, J., Kumar, P., Hodgkinson, A., Luo, H., Dandou, S., Lacroix, M., Dupré, P., Pignodel, C., Larive, R., Le Cam, L., Radulescu, O., Racocceanu, D. (2022). 3D reconstruction and mathematical modelling of whole slide images to elucidate resistance to the targeted therapy in melanoma, *ICSB 2022 - The 21st International Conference on Systems Biology*, Berlin, Germany.
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20. Jimenez Garay G., Kar A., Ounissi M., Stimmer L., Delatour B., Racocceanu D. (2022). Interpretable Deep Learning in Computational Histopathology for refined identification of Alzheimer's Disease biomarkers, Alzheimer's Association Int. Conference - AAIC'22, San Diego, USA.
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23. Trujillano F., Gonzalez, J., Saito, C., Flores, A., Racocceanu, D. (2021). Corn Crops Identification using Multispectral Images from Unmanned Aircraft Systems, IEEE IGARSS, International Geoscience and Remote Sensing Symposium, Brussels, Belgium.
24. Huang, C.-H., Racocceanu, D. (2020), Enhanced Methods for Lymphocyte Detection and Segmentation on H&E-Stained Images using eXclusive Autoencoders, Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC'20), Montréal, Québec, Canada.
25. Marin, L., Casado, F., Racocceanu, D. (2018), An integrative computational pathology approach to classify prostate cancer. Combining phenotypical, genomic and tumor microenvironment figures using deep learning, European Congress on Digital Pathology, Helsinki, Finland, May 29th – June 1st, 2018.
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46. Racoceanu, D., Content-Based Medical Image Retrieval – the Onco-Media project, TRUMA 2005: International Workshop on Task-Relevant Ubiquitous Media Access, associated with the Third ISERE Meeting: Inter-media Semantic Extraction & Reasoning, 18-20 December 2005, National Taiwan University, Taipei, Taiwan.
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49. Zemouri, R., Racoceanu, D., Zerhouni, N. (2004) Coupling the dynamic memory with the static memory of a neural network for the diagnosis by pattern recognition, DX-2004, 15th International Workshop on Principles of Diagnosis, Carcassonne, France.
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Other production

✓ Collective Scientific Reports

1. Eric Tournié, Jean-Marc Allain, Mathieu Arnoux, Nathalie Bardou, Skandar Basrour, Jean-Marc. Bassat, Christian Bergaud, Xavier Brun, Nathalie Destouches, Aurelian Fatu, David Folio, Antoine Grall, Luca Guidoni, Alexandre Guitton, Pierre-Yves Joubert, Arnaud Landragin, Daniel Racoceanu, Christine Restoin, Gwen Salaün, Roger Serra, Didier Trichet, Rapport d'évaluation de l'unité FEMTO-ST – Franche-Comté Électronique, Mécanique, Thermique et Optiques - Sciences et Technologies, Rapport HCERES 2023. <https://hal.science/hal-04505233>

✓ Scientific reports, submitted papers

1. Mehdi Hamadache, Laura Mouton, David Barriere, Cecile Keller, Carine Chassain, Guilhem Pages, Christophe Delhomme, Annick Bernalier, Jean-Marie Bonny, Daniel Racoceanu. Multimodal integration of data characterizing the evolution of the gutbrain axis during the prodromal phase of Parkinson's disease in a rat model. 2025. <https://hal.science/hal-05506190v1>

✓ Patents and copyrights

Patent / Software (Title & Reference)	Details & Role
Virtual staining – paired images <i>Device and method for generating n virtual IHC stain images from one H&E image (paired img)</i> EP 24 305 221.4 (Europe, filed 9 Feb 2024) PCT/EP2025/053151 (World filed 6 Feb 2025) WO2025/168731 (World published 14 Aug 2025)	Field: Computational pathology, virtual immunohistochemistry (IHC) Applicants: ICM, AP-HP, CNRS, INSERM, Sorbonne Univ., Univ. Paris Cité Role: Co-inventor & institutional co-owner
Virtual staining – unpaired images <i>Device and method for generating n virtual IHC stain images from one H&E image (unpaired img)</i> EP 24 305 224.8 (Europe, filed 9 Feb 2024) PCT/EP2025/053153 (World filed 6 Feb 2025) WO2025/168729 (World published 14 Aug 2025)	Field: Computational pathology, virtual IHC Applicants: ICM, AP-HP, CNRS, INSERM, Sorbonne Univ., Univ. Paris Cité Role: co-Inventor & institutional co-owner
3D Tracking in Microscopy <i>Method and system for tracking motion of microscopic objects within a 3-D volume</i> US 2014/0192178 A1, filed 13 Aug 2012, pub. 10 Jul 2014 (A*STAR Singapore)	Field: Biomedical imaging, 3-D cell tracking Role: co-Inventor (holder A*STAR)
Dynamic Real-Time Monitoring System INPI France 0858450, filed 10 Dec 2008 Reference: INPI ref. 8E50BTFR1-Em@systec (Em@systec France)	Field: Industrial vision, sensor systems Role: Co-inventor (holder Em@systec France - spin-off UFC)
Mitosis Detector CNRS DL 05963-01 (UMI 2955 IPAL, 2013)	Field: Computational pathology, cell event detection Role: co-Author (CNRS copyright)
PDFibAtI@s CNRS DL 04137-01 (UMI 2955, submitted 16 Dec 2010, approved 13 May 2011)	Field: Neuroimaging, MRI-DTI data fusion for Parkinson's disease Role: co-Author (CNRS copyright)
HISTOGRAD CNRS DL 2944-01 (UMI CNRS 2955); APPIDN.fr.001.190019.000.S.P.2010.000.20900 (14 May 2010)	Field: Digital pathology, breast cancer grading Role: co-Author (CNRS copyright)

✓ Regional, National and International Disseminations

1. Sarbout, I., Cazenave-coupet, T., Ounissi, M., Milea, D., Racocceanu, D. (2026). City of Light (COL): A City-Scale, Geo-Anchored Urban Simulator with High-Throughput Multi-Sensor Streams, Apidays Paris 2025.
2. Working group of the Inria evaluation committee (2025). Reflections on the Use of Generative AI for Research Professions. <https://inria.hal.science/hal-05188001>. Réflexions sur l'usage de l'IA générative pour les métiers de la recherche. <https://inria.hal.science/hal-05187992>.
3. Ilias Sarbout, Dan Milea, Daniel Racocceanu - Deep Learning for Navigation of the Visually Impaired using Synthetic Data from Blind Digital Twins, Journées Scientifiques Inria "Handicap, perte d'autonomie & numérique", 19-20 Nov. 2024, Paris ; France.
4. Racocceanu, D., Singapore: Asia's Leading Location for Biomedical and Cosmetic Industries, Guest Editor's Note - FCCS Focus Magazine, French Chamber of Commerce, Singapore, Jan. 2014, pp. 3.
5. IPAL: Four more years for the French-Singaporean lab IPAL, CNRS international magazine, no 22, 2011.
6. Le laboratoire franco-singapourien poursuit son essor, Journal CNRS, no 256, May 2011.
7. IPAL – success stories showcase (2011) of the SECAS EU project funded by the European Commission under the FP7 ICT program; Eutema, a research consultancy from Austria, coordinates SECAS. Other partners include Optimat (UK), National ICT Australia (NICTA) and Singapore Management Univ. (SMU).
8. IPAL: A Decade-Long Success Story, FCCS Focus Magazine, French Chamber of Commerce, Singapore, a Living Lab, summer 2010, pp. 22,23.

9. Singapore, New Research Hub, Around the world, Horizons, CNRS International Magazine n° 18 July 2010, pp. 34

✓ **PhD supervision**

1. M. Mehdi OUNISSI, Decoding the Black Box: Enhancing Interpretability and Trust in AI for Biomedical Imaging—A Step Towards Responsible Artificial Intelligence, Sorbonne University, ED130 (EDITE) Doctoral School - Computer Science, Telecommunication And Electronics, Oct. 2024.
2. M. Gabriel Alexandro JIMENEZ GARAY, Representation Learning and Data-Centric Approaches in Computational Pathology. Instantiation to Alzheimer's Disease, Sorbonne University, ED130 (EDITE) Doctoral School - Computer Science, Telecommunication and Electronics, Sept. 2024.
3. Ms. Laura MARIN, Integrative Digital Pathology for Personalized Medicine: Population stratification and early biomarkers findings, consolidating and completing the use of Prostate-Specific Antigen (PSA) and Gleason Score in Prostate Cancer, *Université Pontificale Catholique du Pérou (PUCP)*, 2023.
4. Ms. Oumeima LAIFA, A joint Discriminative-Generative approach for tumour angiogenesis assessment in Computational Pathology, Sorbonne University, co-supervision with Dr Hwee Kuan LEE, BII/A*STAR Singapore, Sept. 2019.
5. M. Lamine TRAORE, Semantic Modelling of a Histopathology Image Exploration and Analysis Tool, PhD of the University Pierre and Marie Curie, ED393 Doctoral School - Public Health: epidemiology and biomedical informatics, Dec. 2017.
6. M. Bassem BEN CHEIKH, Graph-based Mathematical Morphology for the Characterization of the Spatial Organization of Histological Structures in High-Content Images: Application to Tumor Microenvironment in Breast Cancer, University Pierre and Marie Curie, EDITE ED130 Doctoral School - Computer Science, Telecommunication and Electronics, Sept 2017.
7. M. Olivier MORERE, Deep Learning Compact and Invariant Image Descriptors for Instance Retrieval, PhD of the University Pierre and Marie Curie, EDITE, ED130 Doctoral School - Computer Science, Telecommunication and Electronics, June 2016.
8. Ms. Sreetama BASU, Automatic Analysis of Neuronal Morphology: Detection, Modeling and Reconstruction, NUS PhD of the National University of Singapore, March 2015.
9. M. Antoine FAGETTE, Dense Crowd Analysis, PhD of the University Pierre and Marie Curie, EDITE Doctoral School Computer Science, Telecommunication Electronics, June 2014.
10. M. Stéphane RIGAUD, Analysis-Synthesis Approach for Neurosphere Modelisation Under Phase-Contrast Microscopy, PhD of the University Pierre and Marie Curie (EDITE, ED130 Doctoral School - Computer Science, Telecommunication and Electronics), March 2014.
11. M. Humayun IRSHAD, Automated Mitosis Detection in Color and Multi-spectral High-Content Images in Histopathology: Application to Breast Cancer Grading in Digital Pathology, PhD of University Joseph Fourier – Grenoble 1, Jan 2014.
12. M. Antoine VELLARD, Kernel Methods for the Incorporation of Prior-Knowledge into Support Vector Machines, PhD of the National University of Singapore, Dec. 2012.
13. Ms. Roxana Oana TEODORESCU, Parkinson's Disease Prognosis using Diffusion Tensor Imaging Features Fusion / Pronostic de la maladie de Parkinson basé sur la fusion des caractéristiques d'Images par Résonance Magnétique de Diffusion, joint PhD of the University of Franche-Comté and Politehnica University of Timisoara, Apr. 2011.
14. Ms. Adina Eunice TUTAC (ép. BRANICI), Formal Representation and Reasoning for Microscopic Medical Image-Based Prognosis. Application to Breast Cancer Grading / Représentation et Raisonnement Formels pour le Pronostic basé sur l'Imagerie Médicale Microscopique. Application à la Graduation du Cancer du Sein, joint PhD of the University of Franche-Comté and Politehnica University of Timisoara, Timisoara, Oct. 2010.
15. Ms. Eugenia MINCA, Surveillance des systèmes de production en utilisant les réseaux de Petri flous. Application à la E-maintenance des systèmes flexibles de production, joint PhD of the University of Franche-Comté and Univ. Valahia Targoviste, Sept. 2004.
16. M. Nicolas PALLUAT, Méthodologie de surveillance dynamique à l'aide des réseaux neuro-flous temporels, University of Franche-Comté, Jan. 2006.
17. M. Ryad ZEMOURI, Contribution à la surveillance des systèmes à l'aide des réseaux de neurones dynamiques : Application à la e-maintenance, University Franche-Comté, Nov. 2003.

✓ **Habilitation to Supervise Research (HDR – Habilitations à Diriger des Recherches)**

Loménie, N. Habilitation (Habilitation à Diriger des Recherches), Structural and Geometric Analysis of Radiometric Images by Visual Perception Modeling with application to digital pathology, 11th of June 2013, Paris. Jury: Stuart Russel (Univ. Berkeley), Catherine Garbay (CNRS, Univ. Grenoble), Isabelle Bloch (Telecom ParisTech), Laurent Najman (ESIEE & Univ. Marne-la-Vallée), Valérie Gouet-Brunet (Inst. Géographique National), Olivier Lézoray (Univ. Caen), Laurent Wendling (Univ. Paris Descartes), Racoceanu, D. – co-garant (Univ. Pierre Marie Curie), Georges Stamon - garant (Univ. Paris Descartes).

✓ **International Conferences, Symposiums, Special Sessions and Workshops organization**

10. MICCAI 2022, Singapore, Organization Committee (2800 participants)
11. MICCAI 2021, Strasbourg, France, Organization Committee (2700 participants)
12. MICCAI 2020, Lima, Peru – Latin America, **General Chair** (2600 participants)
13. ECDP 2018, European Conf. of Digital Pathology, Helsinki, Finland, Scientific committee.
14. First University of Waterloo - Sorbonne University seminar: " Crystalizing Transdisciplinary innovation in Health Engineering ", 9-11 May, 2016.
15. MITOS & ATYPIA @ ICPR 2014, International Medical Benchmark entitled: Mitosis Detection and Atypia assessment in Breast Cancer Histological Images, in the framework of ICPR 2014, 22nd International Conf. Pattern Recognition, 24-28 Aug 2014, Stockholm, Sweden.
16. ECDP 2014, European Conference of Digital Pathology, Scientific and Organization Committees, 18-20 June 2014, Paris.
17. IJCNN 2013 Special Session: "Computational Intelligence in Bioimage Informatics" (IJCNN-CIBI2013), International Joint Conference on Neural Networks, Dallas, TX, USA, August 4-9, 2013
18. SinFra 2012 - Singaporean-French IPAL Symposium, 15-16 October 2012, University Pierre and Marie Curie and Institut Mines-Telecom, Paris - Scientific invited Sessions, Racoceanu, D., General Chair.
19. MITOS @ ICPR 2012, International Medical Benchmark entitled: Mitosis Detection in Breast Cancer Histological Images, initiative of IPAL in collaboration with the SME TRIBVN (Châtillon, France), Hôpital de la Pitié-Salpêtrière (Paris, France) and Ohio State University (US), in the framework of ICPR 2012, 21st International Conference on Pattern Recognition, 11-15 Nov., 2012, Tsukuba, Japan.
20. IJCNN 2011 Special Session "Consciousness-driven vision: toward a breakthrough in bio-inspired computer vision", 31 July – 5 Aug 2011, US.
21. SFBI 2011 – Singaporean - French Biomedicine Seminar 2011, 24-25 February. 2011, Biopolis, Singapore (Keynote – Mathias Fink, ESPC, Paris).
22. ICT'2010, EU-ASEAN Cooperation opportunities, Networking Session jointly organized by IPAL and SEACOP, Bruxelles, Belgium, 27-29 Sept. 2010.
23. Image Understanding - an important step toward Visual Cognition, 2010 IEEE World Congress on Computational Intelligence (WCCI 2010), Special Session, Chairs: Racoceanu, D. and Huang, C. H., 18-23 July, 2010, Barcelona, Spain.
24. SinFra'09, Singaporean-French IPAL Symposium, 18–20 Feb. 2009, Singapore, - Tutorials & Scientific Sessions, Racoceanu, D., Lim, J. H., Leow, W. K., General Chairs (Keynote Joseph Sifakis, 2007 ACM Turing award winner)
25. "Ontology and Context Related Medical Image Distributed Intelligent Access" - ONCO-MEDIA special session at the JAMIT (Japanese Society of Medical Imaging Technology) Annual Meeting, 20th of July 2007, Tsukuba, Japan, chairs: Nakai, T. & Racoceanu, D.
26. Special Session - Medical Image Retrieval - AIRS 2006 – Asian Information Retrieval Symposium, 18 Oct. 2006, Singapore, General Chair - Racoceanu, D., publications issued in Lecture Notes for Computer Science (LNCS)

27. SFBI'06, 1st Singaporean-French Biomedical Imaging workshop, 12-13 October 2006, Biopolis, Singapore, Organizing Committee Chairs: Racocceanu, D., Benoît-Cattin, H., organized with SBIC (Singapore Biomed Imaging Consortium) and CREATIS/INSA Lyon.